

Form Number : _____

Paper Code
(1201CMD302119015)

* 0000CMD302118004 *



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2019 - 2020)

PRE-MEDICAL : ENTHUSIAST COURSE

PHASE-I

Test Type : S.C.T. # 01

Test Pattern : NEET(UG)

TEST DATE : 05 - 11 - 2019

SYLLABUS COMPLETION TEST

Important Instructions

Do not open this Test Booklet until you are asked to do so

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is 3 Hours and Questions Paper Contains 180 Questions. The Max. Marks are 720.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while 1 mark will be deducted for every wrong answer. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of Blue / Black Ball Point Pen only as the correct answer(s) of the question attempted.
8. Use of Pencil is strictly prohibited.

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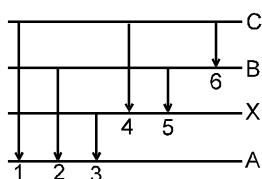
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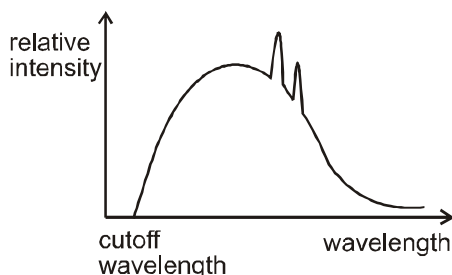
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1. Consider atoms H, He^+ , Li^{++} in their ground states. Suppose E_1 , E_2 and E_3 are minimum energies required so that the atoms H, He^+ , Li^{++} can achieve their first excited states respectively, then-
- (1) $E_1 = E_2 = E_3$ (2) $E_1 > E_2 > E_3$
(3) $E_1 < E_2 < E_3$ (4) $E_1 < E_2 = E_3$
2. In the figure six lines of emission spectrum are shown. Which of them will be absent in the absorption spectrum.

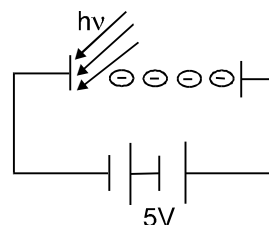


- (1) 1, 2, 3 (2) 1, 4, 6
(3) 4, 5, 6 (4) 1, 2, 3, 4, 5, 6
3. A monochromatic radiation of wavelength λ is incident on a sample containing He^+ . As a result the Helium sample starts radiating. A part of this radiation is allowed to pass through a sample of atomic hydrogen gas in ground state. It is noticed that the hydrogen sample has started emitting electrons whose maximum Kinetic Energy is 37.4 eV. ($hc = 12400 \text{ eV \AA}$) Then λ is-
- (1) 275 $\text{\AA}$ (2) 243 $\text{\AA}$ (3) 656 $\text{\AA}$ (4) 386 $\text{\AA}$
4. A beam of electrons striking a copper target produces X-rays. Its spectrum is as shown. Keeping the voltage same if the copper target is replaced with a different metal, the cut-off wavelength and characteristic lines of the new spectrum will change in comparison with old as :

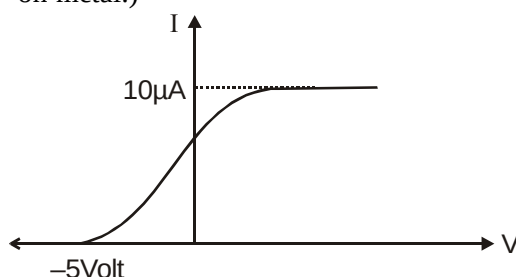


- (1) Cut-off wavelength will remain unchanged while characteristic lines will be different.
(2) Both cut-off wavelength and characteristic lines will remain unchanged.
(3) Both cut-off wavelength and characteristic lines will be different.
(4) Cut-off wavelength will be different while characteristic lines will remain unchanged.

5. Photons of energy 5 eV are incident on cathode. Electrons reaching the anode have kinetic energies varying from 6eV to 8eV. The work function of the metal is :



- (1) 1eV (2) 2eV
(3) 3eV (4) 4eV
6. In the photoelectric experiment, if we use a monochromatic light, the I - V curve is as shown. If work function of the metal is 2eV, estimate the power of light used. (Assume efficiency of photo emission = $10^{-3}\%$, i.e. number of photoelectrons emitted are $10^{-3}\%$ of number of photons incident on metal.)



- (1) 2 W (2) 5 W
(3) 7 W (4) 10 W
7. In an α -decay the Kinetic energy of α particle is 48 MeV and Q-value of the reaction is 50 MeV. The mass number of the mother nucleus is:- (Assume that daughter nucleus is in ground state)
- (1) 96 (2) 100
(3) 104 (4) none of these
8. A heavy nucleus having mass number 200 gets disintegrated into two small fragments of mass number 80 and 120. If binding energy per nucleon for parent atom is 6.5 MeV and for daughter nuclei is 7 MeV and 8 MeV respectively, then the energy released in the decay will be:
- (1) 200 MeV (2) - 220 MeV
(3) 220 MeV (4) 180 MeV

9. Consider the following statements :
- Nuclear fission is normally followed by emission of β^- -particles.
 - Emission of α -particle is normally followed by emission of γ -rays.
 - In carbon-carbon cycle of fusion reaction which powers the large stars, two carbon combine nuclei to form a magnesium nucleus:
- The correct order of True / False in above statements is

- (1) T T T (2) T T F
(3) F T T (4) T F T

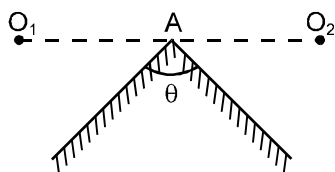
10. Two identical samples (same material and same amount) P and Q of a radioactive substance having mean life T are observed to have activities A_p & A_Q respectively at the time of observation. If P is older than Q, then the difference in their ages is :

- (1) $T \ln \left(\frac{A_p}{A_Q} \right)$ (2) $T \ln \left(\frac{A_Q}{A_p} \right)$
(3) $\frac{1}{T} \ln \left(\frac{A_p}{A_Q} \right)$ (4) $T \left(\frac{A_p}{A_Q} \right)$

11. Nuclei of radioactive element A are produced at rate ' t^2 ' at any time t. The element A has decay constant λ . Let N be the number of nuclei of element A at any time t. At time $t = t_0$, $\frac{dN}{dt}$ is minimum. Then the number of nuclei of element A at time $t = t_0$ is

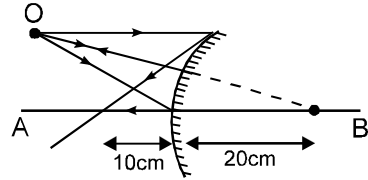
- (1) $\frac{\lambda t_0^2 - 2t_0}{\lambda^2}$ (2) $\frac{t_0 - \lambda t_0^2}{\lambda^2}$
(3) $\frac{2t_0 - \lambda t_0^2}{\lambda}$ (4) $\frac{t_0 - \lambda t_0^2}{\lambda}$

12. Two plane mirrors are joined together as shown in figure. Two point objects O_1 and O_2 are placed symmetrically such that $AO_1 = AO_2$. The image of the two objects is common if :



- (1) $\theta = 60^\circ$ (2) $\theta = 90^\circ$
(3) $\theta = 30^\circ$ (4) $\theta = 45^\circ$

13. In the figure shown a convex mirror of radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the ray diagram (AB is its principal axis):

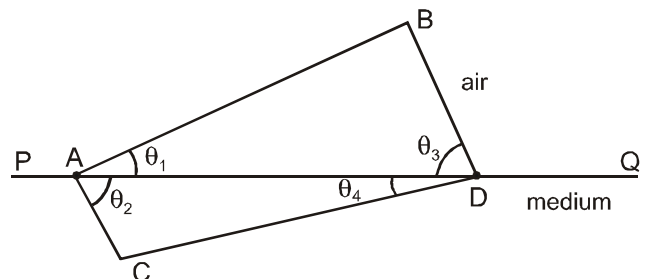


- (1) 3 (2) 2 (3) 1 (4) 0

14. A prism with refracting angle 60° has an angle of minimum deviation 60° in air. Find the angle of minimum deviation if the prism is put inside a liquid of refractive index $\frac{5}{2\sqrt{3}}$:

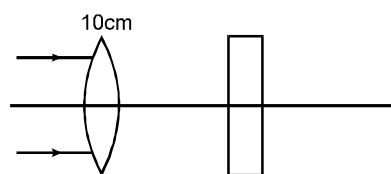
- (1) 14° (2) 41° (3) 24° (4) 28°

15. The shown figure represents a wave front AB which passes from air to another transparent medium and produces a new wave front CD after refraction. The refractive index of the medium is (PQ is the boundary between air and the medium).



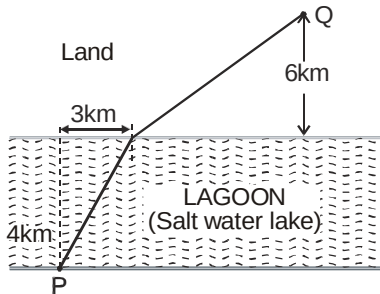
- (1) $\frac{\cos \theta_1}{\cos \theta_4}$ (2) $\frac{\cos \theta_4}{\cos \theta_1}$
(3) $\frac{\sin \theta_1}{\sin \theta_4}$ (4) $\frac{\sin \theta_2}{\sin \theta_3}$

16. A parallel beam of light is incident on a lens of focal length 10 cm. A parallel slab of refractive index 1.5 and thickness 3 cm is placed on the other side of the lens. Find the distance of the final image from the lens.

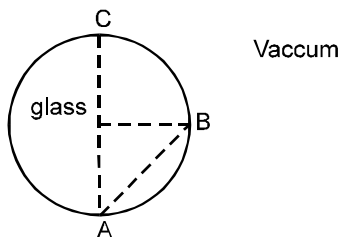


- (1) 15 cm (2) 11 cm (3) 9 cm (4) 8 cm

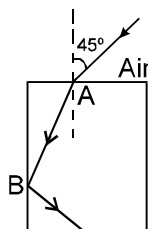
17. A man starting from point P crosses a 4 km wide lagoon and reaches point Q in the shortest possible time by the path shown. If the person swims at a speed of 3 km/hr and walks at a speed of 4 km/hr, then his time of journey is :



- (1) 200 min (2) 250 min
(3) 280 min (4) 300 min
18. It is found that all electromagnetic signals sent from A towards B reach point C. The speed of electromagnetic signals in glass can not be :



- (1) 1.0×10^8 m/s (2) 2.4×10^8 m/s
(3) 2×10^7 m/s (4) 4×10^7 m/s
19. A ray of light is incident from air to a glass rod at point A the angle of incidence being 45° . The minimum value of refractive index of the material of the rod so that T.I.R. takes place at B is _____.

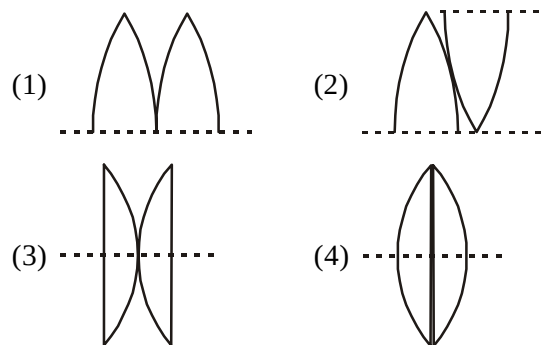


- (1) $\sqrt{\frac{2}{3}}$ (2) $\sqrt{\frac{3}{5}}$
(3) $\sqrt{\frac{3}{2}}$ (4) $\frac{3}{2}$

20. The magnitude of focal length of a plano-convex lens is 10 cm, then its magnitude of focal length when its plane surface is polished (silvered) is :

- (1) 20 cm (2) 5 cm
(3) 10 cm (4) none of these

21. A convex lens is cut into two parts in different ways that are arranged in four manners, as shown. Which arrangement will give maximum optical power ?



22. The following data are given for a crown glass prism ;

refractive index for blue light $n_b = 1.521$

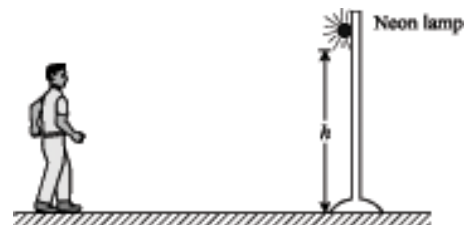
refractive index for red light $n_r = 1.510$

refractive index for yellow light $n_y = 1.550$

Dispersive power of a parallel glass slab made of the same material is :

- (1) 0.01 (2) 0.02
(3) 0.03 (4) 0

23. On a very hot day, a boy standing at a large distance observes a neon monochromatic lamp hanging at a height h above the ground. He observe the lamp as shown in the figure :

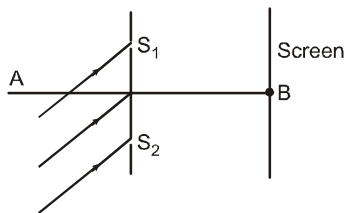


- (1) at a height h .
(2) at a height more than h .
(3) at a height less than h .
(4) at a height that is varying with time, sometimes more than h and sometimes less than h .

- 24.. The wavefront of a light beam is given by the equation $x + 2y + 3z = c$, (where c is arbitrary constant) then the angle made by the direction of light with the y -axis is :

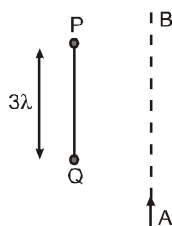
- (1) $\cos^{-1} \frac{1}{\sqrt{14}}$
(2) $\sin^{-1} \frac{2}{\sqrt{14}}$
(3) $\cos^{-1} \frac{2}{\sqrt{14}}$
(4) $\sin^{-1} \frac{3}{\sqrt{14}}$

25. A beam of light parallel to central line AB is incident on the plane of slits in a YDSE experiment as shown. The number of minima obtained on the large screen is n_1 . Now if the beam is tilted by some angle ($\neq 90^\circ$) as shown in the figure, then the number of minima obtained is n_2 . Then:



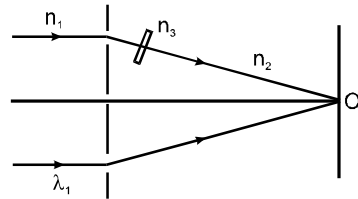
- (1) $n_1 = n_2$ (2) $n_1 > n_2$
(3) $n_2 > n_1$ (4) none of these

26. Two coherent light sources P and Q each of wavelength λ are separated by a distance 3λ as shown. The maximum number of minima formed on line AB which runs from $-\infty$ to $+\infty$ is :

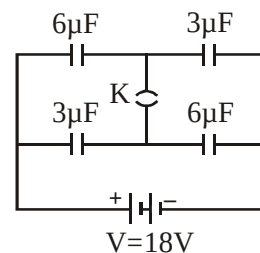


- (1) 2 (2) 4
(3) 6 (4) 8

27. In the figure shown in a YDSE, a parallel beam of light is incident on the slits from a medium of refractive index n_1 . The wavelength of light in this medium is λ_1 . A transparent slab of thickness ' t ' and refractive index is put in front of one slit. The medium between the screen and the plane of the slits is n_2 . The phase difference between the light waves reaching point 'O' (symmetrical, relative to the slits) is :



- (1) $\frac{2\pi}{n_1 \lambda_1} (n_3 - n_2)t$ (2) $\frac{2\pi}{\lambda_1} (n_3 - n_2)t$
(3) $\frac{2\pi n_1}{n_2 \lambda_1} \left(\frac{n_3}{n_2} - 1 \right) t$ (4) $\frac{2\pi n_1}{\lambda_1} (n_3 - n_2) t$
28. Two identically charged pith balls are suspended from the same point by two massless identical threads density of each ball is ρ . If system is immersed in a medium of density σ , balls remain undeflected, then the dielectric constant of medium is :-
- (1) $\frac{\rho}{\rho - \sigma}$ (2) $\frac{\rho - \sigma}{\rho}$ (3) $\frac{\sigma}{\rho - \sigma}$ (4) $\frac{\rho - \sigma}{\sigma}$
29. An electron is released on the axis of a ring having charge Q and radius R from distance $\sqrt{3}R$ from the center. The speed of electron at the center will be. (m is mass of electron)
- (1) $\sqrt{\frac{1}{4\pi\epsilon_0} \frac{Qe}{mR}}$ (2) $\sqrt{\frac{1}{4\pi\epsilon_0} \frac{Qe}{2mR}}$
(3) $\sqrt{\frac{1}{4\pi\epsilon_0} \frac{2Qe}{mR}}$ (4) $\sqrt{\frac{1}{4\pi\epsilon_0} \frac{Qe}{R\sqrt{2}}}$
30. In the given network of capacitors how much charge will flow through the key when it is plugged.



- (1) 27 μC (2) 54 μC (3) 18 μC (4) 9 μC

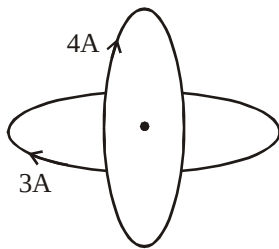
31. Two capacitors of capacitance $600 \mu\text{F}$ and $1200 \mu\text{F}$ can work upto maximum potential of 4 kV and 3 kV respectively. Find maximum potential upto which their series combination can be operated :

(1) 7 kV (2) 9 kV
(3) 6 kV (4) 10 kV

32. Two parallel long wire carrying currents i_1 and i_2 with $i_1 > i_2$. When the currents are in the opposite directions the magnetic field at a point midway between the wires is $90 \mu\text{T}$. If the direction of current in one of the wire is reversed, the field become $30 \mu\text{T}$. The ratio i_1/i_2 is :-

(1) 4 (2) 3 (3) 2 (4) 1

33. Two circular current carrying coils each of radius 10 cm are lying in two perpendicular planes as shown in figure with their centres coinciding. The magnetic field at their common centre is :-



(1) $25\mu_0\text{T}$ (2) $20\mu_0\text{T}$
(3) $15\mu_0\text{T}$ (4) $35\mu_0\text{T}$

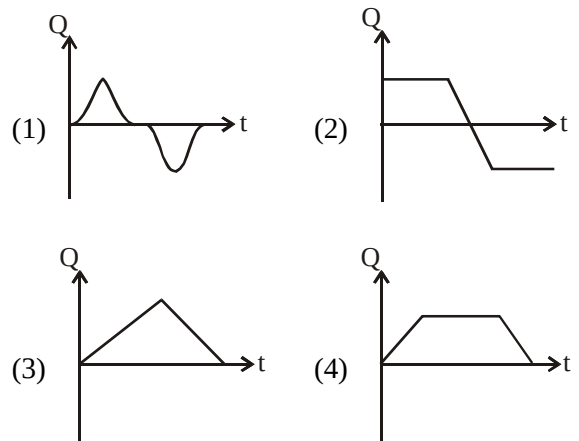
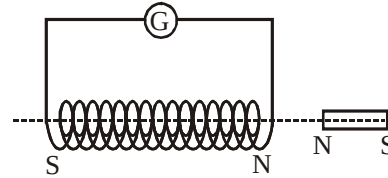
34. If a dip circle is placed in a vertical plane at an angle of 30° to the magnetic meridian, the dip needle makes an angle of 45° with the horizontal the real dip at that place :-

(1) $\tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$ (2) $\tan^{-1} (\sqrt{3})$
(3) $\tan^{-1} \left(\frac{\sqrt{3}}{\sqrt{2}} \right)$ (4) $\tan^{-1} \left(\frac{2}{\sqrt{3}} \right)$

35. A tangent galvanometer has 80 turns of wire. The internal and external diameters of the coil are 19 cm and 21 cm respectively. The reduction factor of the galvanometer at a place where $H = 0.32 \text{ oersted}$ will be ($1 \text{ oersted} = 80 \text{ A/m}$) :-

(1) 0.0064 (2) 0.64
(3) 0.064 (4) None

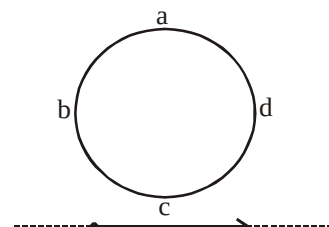
36. A short bar magnet passes at a steady speed right through a long solenoid. A galvanometer is connected across the solenoid. Which graph best represents the variation of the galvanometer deflection θ with time t ?



37. A simple electric motor has an armature resistance of 1Ω and runs from a dc source of 12 V . It draws a current of 2 A when unloaded. When a certain load is connected to it, its speed reduces by 10% of its initial value. The current drawn by the loaded motor is :-

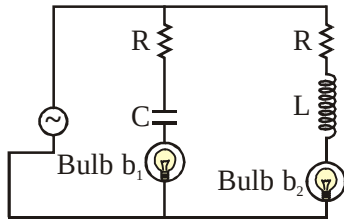
(1) 3 A (2) 6 A
(3) 2 A (4) 1 A

38. An electron moves on a straight line path XY as shown. The $abcd$ is a coil adjacent to the path of electron. What will be the direction of current if any, induced in the will :-



(1) No current induced
(2) $abcd$
(3) $adcb$
(4) Current will reverse its direction as the electron goes past the coil

39. Two identical incandescent light bulbs are connected as shown in the figure. When the circuit is connect to an AC voltage source of frequency f , which of the following observations will be correct ?



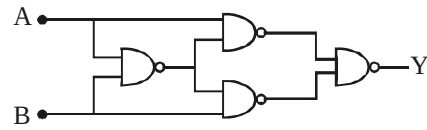
- (1) Both bulbs will glow alternatively
(2) Both bulbs will glow with same brightness provided frequency, $f = \frac{1}{2\pi\sqrt{LC}}$
(3) Bulb b_1 will light up initially and goes off, bulb b_2 will be on constantly
(4) Bulb b_1 will blink and both b_2 will be on constantly

40. An AC source of angular frequency ω is fed across a resistor R and a capacitor C in series. The current registered is I . If now the frequency of source is changed to $\omega/3$ (but maintaining the same voltage), the current in the circuit is found to be halved. Calculate the ratio of reactance to resistance at the original frequency ω .

- (1) $\sqrt{\frac{3}{5}}$ (2) $\sqrt{\frac{2}{5}}$ (3) $\sqrt{\frac{1}{5}}$ (4) $\sqrt{\frac{4}{5}}$

41. Which of the following statements is incorrect?
- (1) In L-C-R series AC circuit, as the frequency of the source increases, the impedance of the circuit first increases and then decreases.
(2) If the net reactance of an L-C-R series AC circuit is same as its resistance, then the current lags/leads behind the voltage by 45° .
(3) At resonance the impedance of an AC circuit becomes purely resistive.
(4) At resonance in L-C-R series AC circuit, the potential drops across inductor and capacitor are equal in magnitude but opposite in phase

42. Truth table for system of four NAND gates as shown in figure is :



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

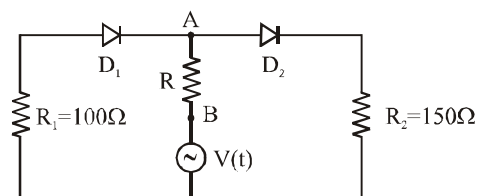
(3)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

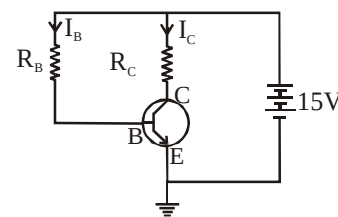
(4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

43. Mobilities of electron and holes in a sample of intrinsic Ge at room temperature are $0.35 \text{ m}^2/\text{V-s}$ and $0.18 \text{ m}^2/\text{V-s}$ respectively. If the electron and hole densities are each equal to $2.5 \times 10^{19}/\text{m}^3$, the Ge conductivity will be :-
(1) 3.12 S/m (2) 2.12 S/m
(3) 1.12 S/m (4) 4.12 S/m
44. In the circuit given below, $V(t)$ is the sinusoidal voltage source voltage drop $V_{AB}(t)$ across the resistance R is:

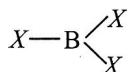


- (1) Is half wave rectified
(2) Is full wave rectified
(3) Has the same peak value in the positive and negative half cycles
(4) Has different peak value during positive and negative half cycle.
45. In the following common emitter circuit if $\beta = 100$, $V_{CE} = 7\text{V}$, $V_{BE} = \text{Negligible}$, $R_C = 2\text{k}\Omega$, then $I_B = ?$



- (1) 0.01 mA (2) 0.04 mA
(3) 0.02 mA (4) 0.03 mA

46. The trend of basicity of lanthanoid hydroxides
 (1) increases across the lanthanoid series
 (2) decreases across the lanthanoid series
 (3) first increases and then decreases
 (4) first decreases and then increases
47. In BX_3 , B – X distance is shorter than what is expected theoretically because (X = F, Cl, Br, I)



- (1) sp^3 hybridisation of B is responsible for shorter B – X distance.
 (2) B – X has a double bond character due to back-bonding.
 (3) Dimerisation takes place in BX_3 which is responsible for shorter B – X distance.
 (4) Due to large size of X, B – X distance decreases.
48. Complete the following reactions by filling the appropriate choice.
- (A) $6XeF_4 + 12H_2O \rightarrow 4Xe + 2XeO_3 + \underline{(i)} + \underline{(ii)}$
 (B) $XeF_6 + 3H_2O \rightarrow \underline{(iii)} + 6HF$
- | | | |
|------------|---------|-----------|
| (i) | (ii) | (iii) |
| (1) F_2 | H_2O | $XeOF_4$ |
| (2) $24HF$ | $3O_2$ | XeO_3 |
| (3) $2HF$ | $2H_2O$ | XeO |
| (4) HF | H_2O | Xe_2O_3 |
49. What is true about S_8 molecule ?
 I. There is no $P_\pi-P_\pi$ bonds present in the molecule
 II. There are eight lone pair of electrons
 III. Each S atom is sp^3 hybridised
 (1) I, III (2) II, III
 (3) I, II (4) only II
50. In which of the following the name of the ore is not matched with its formula?
 (1) Cassiterite – SnO_2
 (2) Limonite – $Fe_2O_3 \cdot 3H_2O$
 (3) Siderite – $FeCO_3$
 (4) Anglesite – $PbCO_3$

51. The correct order of dipole moment is :-
 (1) $CH_4 < NF_3 < NH_3 < H_2O$
 (2) $NF_3 < CH_4 < NH_3 < H_2O$
 (3) $NH_3 < NF_3 < CH_4 < H_2O$
 (4) $H_2O < NH_3 < NF_3 < CH_4$
52. Electronic configuration of Gd(64) is :-
 (1) $[Xe] 4f^2, 5d^9, 6s^2$ (2) $[Xe] 4f^7, 5d^1, 6s^2$
 (3) $[Xe] 4f^6, 5d^2, 6s^2$ (4) $[Xe] 4f^7, 5d^0, 6s^2$
53. Available Cl_2 is liberated from bleaching powder when it :-
 (1) Is heated
 (2) React with excess amount of acid
 (3) React with water
 (4) React with CO_2
54. Which of the following is Acetylide ion :
 (1) C^{2-} (2) C_2^- (3) C_2^{2-} (4) C_3^{4-}
55. $Cl_2 \xrightarrow[\text{dil. NaOH}]{\text{cold and}} X$; X is :-
 (1) $NaClO_3$ (2) $NaOCl$ (3) $NaClO_2$ (4) $NaClO_4$
56. For the reaction, $A + B \longrightarrow C$
- | | | |
|----------|-------|--------------------------|
| [A] | [B] | $-\frac{d[A]}{dt}$ |
| 1. 1.0 M | 1.0 M | 0.25 M min ⁻¹ |
| 2. 2.0 M | 1.0 M | 0.50 M min ⁻¹ |
| 3. 1.0 M | 2.0 M | 0.25 M min ⁻¹ |
- Hence, rate law is
 (1) $k[A][B]$ (2) $k[A][B]$
 (3) $k[A]$ (4) $k[B]$
57. Which of the following is more stable :-
 (1) Cr^{+4} (2) Cr^{+2} (3) Cr^{+3} (4) Cr^{+5}
58. Sometimes it is possible to separate two sulphide ores by adjusting the proportion of oil to water or by using depressants. When a depressant NaCN is added to an ore containing ZnS and PbS, what is the correct observation?
 (1) NaCN prevents PbS from coming to the froth but allows ZnS to come with froth.
 (2) NaCN prevents ZnS from coming to the froth but allows PbS to come with froth.
 (3) NaCN prevents frothing of both ZnS and PbS, hence no froth is formed.
 (4) NaCN does not act as depressant hence a mixture of PbS and ZnS is found in froth.

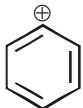
59. The zone refining of metal is based on the principle of :-

- (1) Greater mobility of pure metal than that of impurity
- (2) Higher melting point of the impurity than that of the pure metal.
- (3) Greater noble character of the solid metal than that of the impurity
- (4) Greater solubility of the impurity in the molten state than in the solid

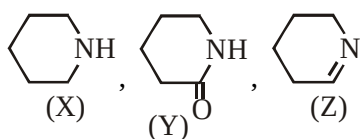
60. Due to the presence of ambidentate ligands coordination compounds show isomerism. Palladium complexes of the type $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{SCN})_2]$ and $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{NCS})_2]$ are :-

- (1) linkage isomers
- (2) coordination isomers
- (3) ionisation isomers
- (4) geometrical isomers

61. Which of the following carbocation is most stable?

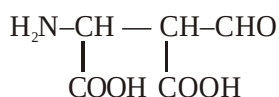
- (1) CH_3CH_2^+
- (2) $\text{CH}_2=\text{CH}^+$
- (3) $\text{CH}\equiv\text{C}^+$
- (4) 

62. Correct order of basic strength for following compounds :-



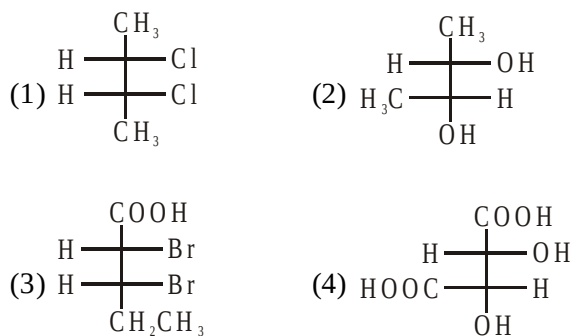
- (1) $X > Y > Z$
- (2) $X > Z > Y$
- (3) $Y > X > Z$
- (4) $Y > Z > X$

63. IUPAC name of following compound ?



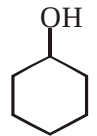
- (1) 3-Amino-2-formylbutane-1,4-dioic acid
- (2) 3-Amino-2,3-dicarboxypropanal
- (3) 2-Amino-3-formyl butane-1,4,-dioic acid
- (4) 1-Amino-2-formyl ethane-1,2-dicarboxylic acid

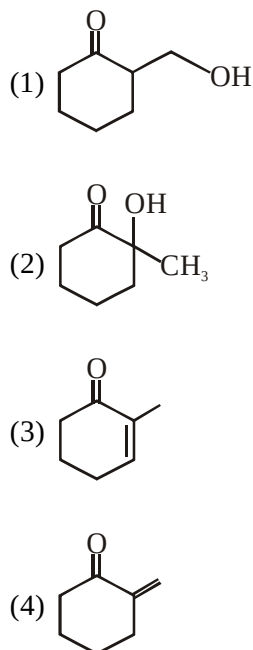
64. Which of the following is chiral :



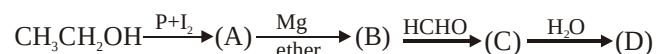
65. Which will show geometrical isomerism :-

- (1) $\text{Ph}-\text{CH}=\text{N}-\text{OH}$
- (2) $\text{CH}_3-\text{N}=\text{N}-\text{C}_2\text{H}_5$
- (3) 
- (4) All

66.  $\xrightarrow{\text{PCC}}$ (A) $\xrightarrow[\Delta]{\text{HCHO/dil. KOH}}$ (B)
 structure of (B) is :-



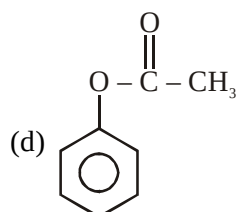
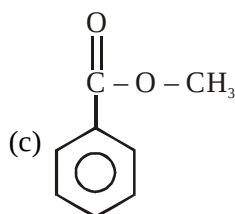
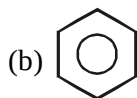
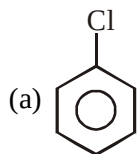
67. In the following sequence of reaction



the compound 'D' is :-

- (1) Propanal
- (2) Butanal
- (3) n-butyl alcohol
- (4) n-propyl alcohol

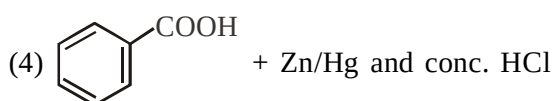
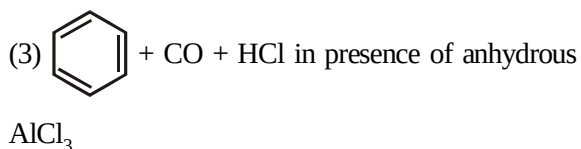
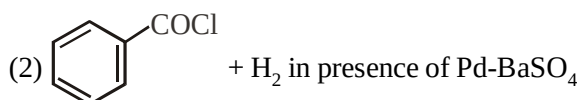
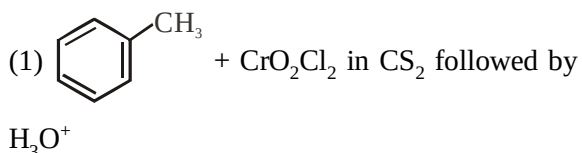
68. Compare the rate of electrophilic substitution reaction?



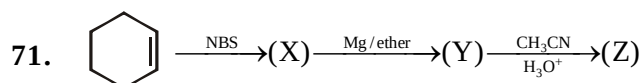
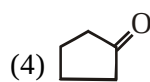
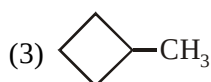
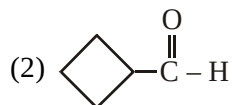
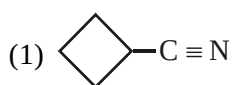
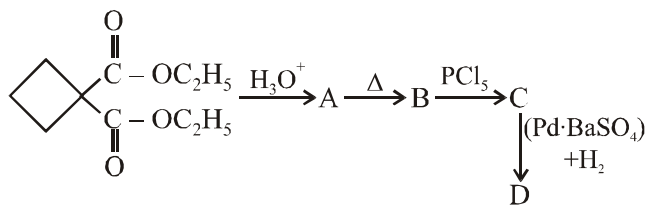
(1) $a > b > c > d$ (2) $b > a > c > d$

(3) $d > b > c > a$ (4) $d > b > a > c$

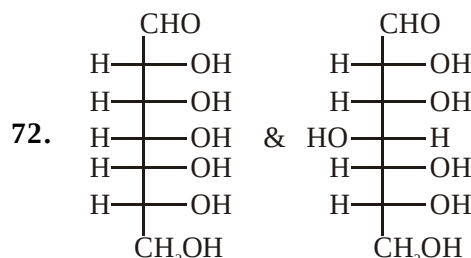
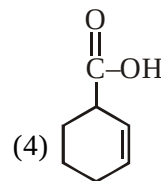
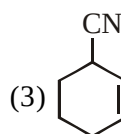
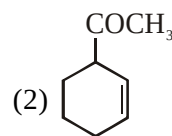
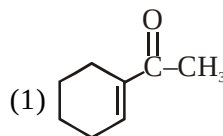
69. Reaction by which benzaldehyde cannot be prepared is :-



70. The end product of the sequence reactions is ?



The product (Z) is :-



are C_n epimers. What is n ?

(1) 3

(2) 4

(3) 2

(4) The given statement is wrong, they are not epimers

73. Nylon is a :-

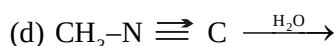
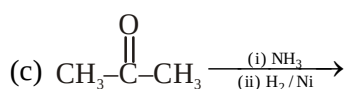
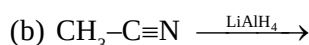
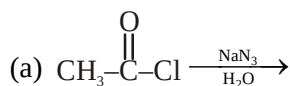
(1) Polyester

(2) Poly propylene

(3) Polyamide

(4) Poly carbonyl compound

74. Which reaction gives primary amines as product:



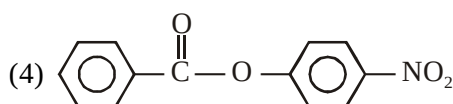
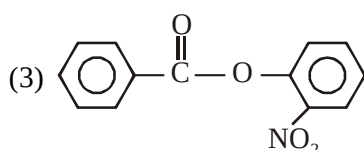
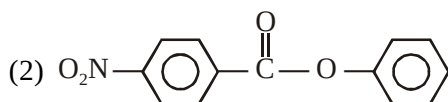
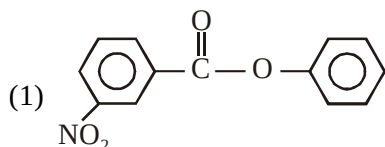
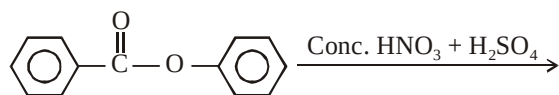
(1) a, b, c, d

(2) a, b, d

(3) b, d

(4) a, b, c

75. The major product in the reaction is :



76. For a reaction $pA + qB \longrightarrow \text{products}$, the rate law expression is $r = k[A]^1[B]^m$, then :

- (1) $(p + 1) < (1 + m)$
- (2) $(p + q) > (1 + m)$
- (3) $(p + q)$ may or may not be equal to $(1 + m)$
- (4) $(p + q) = (1 + m)$

77. Given that :

$$\text{Ni}^{2+} | \text{Ni} = -0.25 \text{ V};$$

$$\text{Cu}^{2+} | \text{Cu} = 0.34 \text{ V}$$

$$\text{Ag}^+ | \text{Ag} = 0.80 \text{ V};$$

$$\text{Zn}^{2+} | \text{Zn} = -0.76 \text{ V}$$

Which of the following reactions under standard condition will not take place in the specified direction:

- (1) $\text{Ni}_{(\text{aq.})}^{2+} + \text{Cu}_{(\text{s})} \longrightarrow \text{Ni}_{(\text{s})} + \text{Cu}_{(\text{aq.})}^{2+}$
- (2) $2\text{Ag}_{(\text{aq.})}^+ + \text{Cu}_{(\text{s})} \longrightarrow 2\text{Ag}_{(\text{s})} + \text{Cu}_{(\text{aq.})}^{2+}$
- (3) $\text{Zn}_{(\text{s})} + \text{Cu}_{(\text{aq.})}^{+2} \longrightarrow \text{Zn}_{(\text{aq.})}^{+2} + \text{Cu}_{(\text{s})}$
- (4) $2\text{H}_{(\text{aq.})}^+ + \text{Zn}_{(\text{s})} \longrightarrow \text{H}_{2(\text{g})} + \text{Zn}_{(\text{aq.})}^{2+}$

78. On heating a mixture of NH_4Cl and KNO_2 , we get

- (1) NH_4NO_3
- (2) $\text{KNH}_4(\text{NO}_3)_2$
- (3) N_2
- (4) NO

79. Atomicity of phosphorus is

- (1) one
- (2) two
- (3) three
- (4) four.

80. When three parts of conc. HCl and one part of conc. HNO_3 is mixed, a compound 'X' is formed. The correct option related to 'X' is

- (1) X is known as aqua-regia
- (2) X is used for dissolving gold
- (3) X is used for decomposition of salts of weaker acids
- (4) both (1) and (2).

81. What will be the freezing point of 0.2 molal aqueous solution of MgBr_2 ? If salt shows 60% dissociation in solution and K_f for water is $1.86^\circ\text{C Kg mol}^{-1}$.

- (1) -0.59°C
- (2) -0.67°C
- (3) -0.82°C
- (4) -1.04°C

82. Xenon has closed shell configuration but is known to give compounds with fluorine because

- (1) Xe atom has large size and lower ionisation potential as compared to other noble gases
- (2) Xe has unpaired electrons which can form covalent bonds
- (3) Xe has highest boiling point hence it can form compounds with fluorine
- (4) fluorine is the smallest element hence it can react with all noble gases.

83. Rearrange the following (I to IV) in the order of increasing masses :-

- (I) 0.5 mole of O_3
- (II) 0.5 g atom of oxygen
- (III) 3.011×10^{23} molecules of O_2
- (IV) 5.6 litre of CO_2 at 1 atm and 273 K

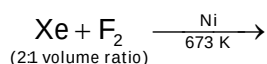
- (1) $II < IV < III < I$
- (2) $II < I < IV < III$
- (3) $IV < II < III < I$
- (4) $I < II < III < IV$

84. Following reaction describes the rusting of iron
 $4Fe + 3O_2 \rightarrow 4Fe^{3+} + 6O^{2-}$

Which one of the following statement is incorrect?

- (1) This is an example of redox reaction
- (2) Metallic iron is reduced to Fe^{3+}
- (3) Fe^{3+} is an oxidising agent
- (4) Metallic iron is a reducing agent

85. Which compound is prepared by the following reaction:



- (1) XeF_4
- (2) XeF_2
- (3) XeF_6
- (4) None of these.

86. An atomic solid crystallizes in a body centre cubic lattice and the inner surface of the atoms at the adjacent corner are separated by 60.3 pm. If the atomic mass of A is 48, then density of the solid, is nearly :-

- (1) 2.7 g/cc
- (2) 5.07 g/cc
- (3) 3.5 g/cc
- (4) 1.75 g/cc

87. A detergent ($C_{12}H_{25}SO_4^-Na^+$) solution becomes a colloidal sol at a concentration of $10^{-3}M$. On an average 10^{13} colloidal particles are present in 1 mm^3 . What is the average number of ions which are contained by one colloidal particle (micelle) ? [Given : $N_A = 6 \times 10^{23}$]

- (1) 6×10^7
- (2) 10
- (3) 60
- (4) None of these

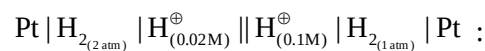
88. Consider the following isomers.

- (i) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$
- (ii) $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$
- (iii) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{NO}_2$

Which of the following observations is correct?

- (1) (i) will give a pale yellow and (ii) will give a white precipitate with AgNO_3 solution.
- (2) (iii) will give a white precipitate with AgNO_3 solution.
- (3) (i), (ii) and (iii) will give white precipitate with AgNO_3 solution.
- (4) None of the above isomers will give white precipitate with AgNO_3 solution.

89. At 298 K the emf of cell is -



- (1) 0.05 V
- (2) 0.0295 V
- (3) 0.1 V
- (4) 0.059 V

90. The number of octahedral void in bcc structure is:-

- (1) 0
- (2) 1
- (3) 2
- (4) 4

- 91.** What is correct when number of chromosomes in leaves of angiosperms is 80 ?
(1) 40 in stem cells (2) 40 in embryo
(3) 30 in endosperm (4) 40 in gametes
- 92.** Pollen grain of a flower is transferred to the stigma of another flower of the same plant. The phenomenon is :-
(1) Genetically self pollination
(2) Ecologically cross pollination
(3) Genetically allogamy
(4) (1) and (2) both
- 93.** During early development of embryosac, the nuclear division is of
(1) Free nuclear type (2) Cellular type
(3) Amitotic type (4) meiotic type
- 94.** More than one additional embryos are formed in polyembryony from :-
(1) Nucellus (2) Integuments
(3) Synergids (4) All of above
- 95.** Pollination by act of "Pseudo copulation" occurs in:-
(1) *Salvia* (2) *Yucca*
(3) *Ophrys* (4) *Nicotia*
- 96.** Which of the following structure is diploid ?
(1) Antipodals (2) Synergids
(3) Secondary nucleus (4) Gamete
- 97.** Insect pollinated flowers have which of the following feature :-
(1) Winged pollen
(2) Large sized pollens
(3) Sticky pollens with rough surface stigma
(4) Small pollens with sticky surface stigma
- 98.** What will be the ploidy of the cells of the nucellus and megaspore ?
(1) n and $2n$ (2) $2n$ and n
(3) n and n (4) $2n$ and $2n$
- 99.** Coconut plant has 24 chromosomes in leaves cells, then what is the number of chromosome in drinking part of coconut ?
(1) 24 (2) 48
(3) 36 (4) 72
- 100.** A tetraploid male plant was crossed with a octaploid female plant, then the ploidy level of the endosperm will be :-
(1) $8n$ (2) $10n$ (3) $7n$ (4) $9n$
- 101.** If an organism is Homozygous for autosomal gene (a) and Heterozygous for gene (B) & (C). What is the probability of gamete whose genotype is ABC.
(1) $\frac{1}{4}$ (2) $\frac{1}{8}$ (3) $\frac{2}{4}$ (4) Zero
- 102.** What is common between Mendelian monohybrid cross and dihybrid cross?
(1) Dominance
(2) Segregation
(3) Independent assortment
(4) More than one options are correct
- 103.** Mutation can produce :-
(1) Incomplete dominant allele
(2) Recessive allele
(3) Co-dominant allele
(4) All of the above
- 104.** Which statement is wrong about Turner's syndrome ?
(1) It is a monosomic disorder
(2) Associated with sex chromosome
(3) Fertile female
(4) Improper development of breast
- 105.** "Segregation of one pair of characters is independent of the other pair of characters". This explains :-
(1) Independent assortment
(2) Law of segregation
(3) Dominance phenomenon
(4) Postulate of paired factors
- 106.** A pure tall plant is planted in soil, which is poor in nutrition and reached the size of dwarf plant. This dwarf plant is crossed with another dwarf plant, the phenotype in F_2 - generation is most likely to be:-
(1) All tall
(2) 50% tall, 50% dwarf
(3) 75% tall, 25% dwarf
(4) Data insufficient
- 107.** What will be the genotypes of parents whose offsprings have equal percentage of four phenotypes ?
(1) $AaBBCCdd \times aabbCCdd$
(2) $AABBCC \times AaBbCc$
(3) $AaBbCCDD \times aabbCCDD$
(4) $AaBbCcDD \times AaBbCcDD$

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- 108.** Consider the cross CcDdEe × ccddEe. What will be the probability of following respectively?
(A) Heterozygous individuals
(B) Homozygous recessive individuals
(C) Individuals homozygous for first character & heterozygous for second and third characters
- (1) $\frac{4}{16}, \frac{1}{4}, \frac{2}{8}$ (2) $\frac{2}{16}, \frac{1}{16}, \frac{1}{8}$
(3) $\frac{1}{8}, \frac{2}{16}, \frac{1}{8}$ (4) $\frac{1}{4}, \frac{1}{16}, \frac{1}{8}$
- 109.** The inheritance of two genetic diseases namely phenylketonuria and haemophilia are to be studied in two different individuals. Both these studies will involve the use of :-
(1) Autosomes
(2) Sex chromosomes
(3) Restriction endonuclease
(4) Pedigree analysis
- 110.** During meiosis-I, homologous chromosomes of one pair switch their position without affecting the position of another pair. This is known as –
(1) Segregation
(2) Separation
(3) Co-dominance
(4) Independent assortment
- 111.** Some gene, which are transcribed, but not translated are those that produce :-
(1) t-RNA and m-RNA
(2) t-RNA and r-RNA
(3) r-RNA and m-RNA
(4) m-RNA
- 112.** The enzyme primase is a :-
(1) DNA dependent, DNA polymerase
(2) RNA dependent, DNA polymerase
(3) DNA dependent, RNA polymerase
(4) RNA dependent, RNA polymerase
- 113.** How many types of amino acyl synthetase enzyme are found in cell ?
(1) 20 types (2) 61 types
(3) 64 types (4) 100 types
- 114.** Guanosine & Cytosine are respectively :-
(1) Nucleotide, Nucleoside
(2) Nucleoside, Nucleotide
(3) Nucleoside, Nucleoside
(4) None of the above
- 115.** hnRNA is formed by which enzyme in prokaryotes?
(1) RNA polymerase I
(2) RNA polymerase II
(3) RNA polymerase III
(4) None of the above
- 116.** Chargaff's equivalency rule is not applicable for which organism ?
(1) *E.coli* (2) Human
(3) Rat (4) TMV
- 117.** If the total amount of adenine in a double stranded DNA is 22.5%, then find out the amount of cytosine(A), Base ratio (B) and kingdom(C) of this DNA sample
(1) A → 27.5%, B → 0.81, C → Prokaryote
(2) A → 25.7%, B → 1.22, C → Eukaryote
(3) A → 27.5%, B → 1.22, C → Protista
(4) A → 27.5%, B → 0.81, C → Monera
- 118.** Unwinding of DNA double helix during replication is carried out by
(1) ATP dependent DNA topoisomerase enzyme
(2) ATP independent DNA helicase enzyme
(3) ATP independent DNA topoisomerase enzyme
(4) ATP dependent DNA helicase enzyme
- 119.** Semiconservation mode of DNA was experimentally proved by using radioactive isotopes, by
(1) Meselson and stahl
(2) Taylor et. al
(3) Temin and baltimore
(4) Hersey and chase
- 120.** Unequivocal proof for biochemical nature of genetic material i.e., DNA, was provided by
(1) Freiderick Meiseher and Altman
(2) Freiderick Griffith
(3) Avery, Macleod and McCarty
(4) Hersey and Chase
- 121.** Capping in hnRNA is catalysed by
(1) Poly A polymerase
(2) SnRNA
(3) Guanyl transferase
(4) Catalytic RNA
- 122.** _____ of the genome codes for protein in human beings.
(1) 98% (2) 50%
(3) 24% (4) < 2%

123. Gause principle is correct when :-

- (a) Resources are unlimited.
- (b) Resources are limited.
- (c) Species are living in same habitat.
- (d) Species are living in different habitat.
- (1) a, c (2) b, c (3) a, d (4) b, d

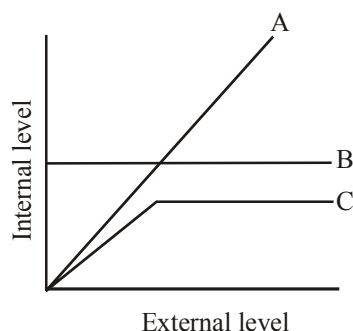
124. Predator are important for a natural ecosystem because :-

- (a) They keep pray population under control.
- (b) Help in maintaining species diversity
- (c) They are used in biological control method.
- (d) They reduce the intensity of competition among competing pray species.
- (1) Only a, c, d (2) Only a, b, c
- (3) Only a, b, d (4) All of the above

125. Which of the following alternatives used by zooplanktons to over come partial stressful conditions ?

- (1) Migration (2) Diapause
- (3) Hibernation (4) Aestivation

126.



What is A, B and C in above diagram?

A B C

- (1) Regulators Conformers Partial regulators
- (2) Regulators Conformers Partial conformers
- (3) Conformers Regulators Partial regulators
- (4) Conformers Regulators Partial conformers

127. A food web :-

- (1) Increases variety of food at each trophic level.
- (2) Imbalances the inter relationship amongst organisms.
- (3) Decreases variety of food, but increases quantity of food at each trophic level.
- (4) Increases variety as well as quantity of food at each trophic level.

128. Limitation of ecological pyramids is/are :-

- (1) it does not take into account the same species belonging to two or more trophic levels
- (2) it assumes a simple food chain, it does not accommodate a food web
- (3) saprophytes are not given any place
- (4) All of the above

129. Standing crop is -

- (i) Biomass in a unit area
- (ii) Number of living organism in unit area
- (iii) Amount of nutrient such as carbon, nitrogen, calcium in unit area
- (iv) Amount of Detritus in unit area
- (1) only (i) (2) (i) & (ii)
- (3) (i), (ii), (iii) (4) (i), (ii), (iv)

130. We have a moral duty to care for Biodiversity, well being and pass on our biological legacy in good order to future generation, this Biodiversity conservation argument is considered in :-

- (1) Narrowly utilitarian
- (2) Ethical argument
- (3) Broadly utilitarian
- (4) All of the above

131. Which one factor is not included in "Evil quartet"?

- (i) Habitat loss and fragmentation
- (ii) Over exploitation
- (iii) Alien species invasion
- (iv) Pollution and disturbance
- (v) Co-extinction
- (1) v (2) iv
- (3) ii (4) iii

132. Which one is not a secondary pollutant ?

- (1) Smog (2) Acid Rain
- (3) PAN (4) DDT

133. Match the items in column I and column II and choose the correct option :-

Column I

Column II

- | | |
|---------------------------------|----------------------|
| A. UV | (i) Biomagnification |
| B. Biodegradable organic matter | (ii) Eutrophication |
| C. DDT | (iii) Snow blindness |
| D. Phosphates | (iv) BOD |

The correct match is :-

- (1) A-ii, B-i, C-iv, D-iii (2) A-iii, B-ii, C-iv, D-i
- (3) A-iii, B-iv, C-i, D-ii (4) A-iii, B-i, C-iv, D-i

134. Root of any plant breeding programme is

- (1) Mutation
- (2) Genetic variability
- (3) Hybridisation
- (4) Selection

135. Which chemical is used in somatic hybridisation ?

- (1) Poly Ethylene glycole
- (2) Acredine
- (3) HNO_2
- (4) Ehtenol

136. Select correct statement w.r.t. reproduction in organisms

- (1) Asexual reproduction involve syngamy
- (2) Haploid-diploid alternation does not occur in asexual reproduction
- (3) Asexual reproduction does not introduce variability and it is slow method of reproduction
- (4) Sexual reproduction involve amitosis

137. Select correct match w.r.t. life span of organism

Column I

Column II

- | | |
|--------------|--------------------|
| a. Parrot | (i) 140 years |
| b. Crow | (ii) 60 years |
| c. Tortoise | (iii) 15 years |
| d. Crocodile | (iv) 100-150 years |

- (1) a(ii), b(iii), c(i), d(iv)
- (2) a(iii), b(ii), c(i), d(iv)
- (3) a(i), b(iii), c(iv), d(ii)
- (4) a(i), b(ii), c(iii), d(iv)

138. Organisms exhibiting external fertilisation are

- (1) Sponges and Cnidarians
- (2) Mammals and Birds
- (3) Reptiles and Sponges
- (4) Echinoderms and Hemichordates

139. The male accessory ducts include

- (1) Tubuli recti, vasa efferentia, epididymis and vas deferens
- (2) Tubuli recti, rete testis, vasa efferentia and vas deferens
- (3) Epididymis and vas deferens only
- (4) Rete testis, vasa efferentia, epididymis and vas deferens

140. Which of the following is incorrect for vas deferens?

- (1) The seminiferous tubules of the testis open into vas deferens through rete testis
- (2) The vas deferens leave the testis and open into epididymis located along the posterior surface of each testis
- (3) They vary from 10 to 20 in number
- (4) All of the above

141. Find out the correct sequence in embryonic development of animal

- (1) cleavage → zygote → fertilization → morula → blastula → gastrula
- (2) fertilization → cleavage → morula → zygote → blastula → gastrula
- (3) fertilization → zygote → blastula → morula → cleavage → gastrula
- (4) fertilization → zygote → cleavage → morula → blastula → gastrula

142. Seminal plasma

- (1) Is not produced by the testes
- (2) Contains fructose which nourishes the sperm
- (3) Contains prostaglandins
- (4) All of the above

143. Which of the following statements regarding the sexual development of the foetus is correct?

- (1) In the absence of a Y chromosome the sexual development of a fetus will follow the male pattern
- (2) The development of the fetal gonads into the male type depends on the presence of oestrogens.
- (3) The sex of an individual is determined by a set of genes on the X chromosome
- (4) The sex of an individual is determined by genes on the Y chromosome

144. Which of the following is incorrect statement?

- (1) Positive feedback by oestrogen cause FSH surge
- (2) Relixin is not secreted by corpus luteum of normal menstrual cycle
- (3) LH surge induce corpus luteum formation
- (4) Placenta is complex structure formed by maternal tissue only

- 145.** Which one of the following is the incorrect matching of the events occur during menstrual cycle ?
- (1) Menstruation : Breakdown of endometrium
 - (2) Ovulation : LH and FSH attain peak level
 - (3) Proliferative phase : Rapid regeneration of endometrium and maturation of Graafian follicle.
 - (4) Development of corpus luteum : Luteal phase and increased secretion of FSH.
- 146.** Just before ovulation in the human female, all the following are true except that :-
- (1) Meiosis-I has just completed
 - (2) Second polar body has just been expelled
 - (3) Meiosis-II is arrested at metaphase
 - (4) Presence of graafian follicle in ovary
- 147.** For prevention of STDs following simple principle should be followed except
- (1) Avoiding sex with multiple/unknown partners
 - (2) Use of condoms
 - (3) If STDs is detected one should get isolated from society
 - (4) The infected person must consult a qualified doctor
- 148.** Given below are four statements (a-d) regarding assisted reproductive technologies:-
- (a) ZIFT - The zygote or embryo (with upto 8 cells) transferred into the fallopian tube
 - (b) ICSI - A sperm is directly injected into the ovum to form an embryo in the laboratory
 - (c) AI - The semen collected either from the husband or healthy donor is artificially introduced into the uterus
 - (d) GIFT - Transfer of ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilisation and development
- Which two of the above statements are correct:
- (1) a and d
 - (2) b & c
 - (3) a & c
 - (4) a, b, c & d

- 149.** Which of the following statement is incorrect?
- (1) MTPs are considered relatively safe during the first month
 - (2) Pills are very effective with less side effects and are well accepted by the females
 - (3) Natural methods of contraception work on the principle of preventing physical meeting of ovum and sperms
 - (4) Surgical method of contraception prevents gamete transport
- 150.** Which of the following contraceptive methods correctly matches with its mode of action?
- | Contraceptive method | Mode of action |
|-----------------------------|--|
| (1) Vasectomy | Make the uterus unsuitable for implantation |
| (2) Condoms | Inhibit ovulation and implantation |
| (3) Diaphragm | Spermicidal and increases phagocytosis of sperms within the uterus |
| (4) IUDs | Increase phagocytosis of sperm |
- 151.** Acid in the stomach, saliva in the mouth, lysozyme in tears, all prevent microbial growth. These all are included in :-
- (1) Non-specific immunity
 - (2) Specific immunity
 - (3) Physiological barrier
 - (4) (1) and (3) Both
- 152.** Match Column -I with Column-II and select the correct answer from codes given below.
- | Column-I | Column-II |
|-----------------|--|
| (A) Sporozoites | (i) Infectious form of <i>Plasmodium</i> |
| (B) Filariasis | (ii) <i>Aedes</i> |
| (C) Typhoid | (iii) <i>Wuchereria</i> |
| (D) Chikungunya | (iv) Widal test |
- (1) A-(iv), B-(ii), C-(i), D-(iii)
 - (2) A-(iii), B-(iv), C-(ii), D-(i)
 - (3) A-(i), B-(iii), C-(iv), D-(ii)
 - (4) A-(ii), B-(iii), C-(i), D-(iv)

153. Which one of the following immune system components does not correctly match with its respective role?

- (A) **Interferons:** secreted by virus-infected cells and protect virus-infected cells from further viral infection
(B) **Plasma cells:** Produce antibodies in response to pathogens into blood to fight with them
(C) **Macrophages:** Cytokine secreting cells that phagocytize microbes entering in the body
(D) **IgA:** Present in colostrum in early days of lactation to protect fetus from diseases

- (1) A & D (2) A & C
(3) B & D (4) Only C

154. How many of the following diseases can be controlled by eliminating the vectors and their breeding places

- i. Dengue ii. Chikungunya
iii. Ascariasis iv. Filariasis
v. Typhoid vi. Bubonic plague
(1) two (2) three
(3) four (4) five

155. Which one of the following options gives the correct matching of a disease with its causative organism and mode of infection

	Disease	Causative Organisms	Mode of Infection
(1)	Elephantiasis	<i>Wuchereria bancrofti</i>	Contaminated water and food
(2)	Malaria	<i>Plasmodium vivax</i>	Biting of male Anopheles mosquito
(3)	Typhoid	<i>Salmonella typhi</i>	contaminated air
(4)	Pneumonia	<i>Streptococcus Pneumoniae</i>	Droplet infection

156. Metastasis is

- (1) Transformation of normal cells to neoplastic cells
(2) When cancerous cells remain confined to their original location
(3) The movement of cancerous cells to distant sites through blood and lymph
(4) When normal cells starve and die due to the lack of nutrients

157. Which of the following is characteristic feature of cancerous/neoplastic cells?

- (1) Exhibit contact inhibition
(2) Starve the normal cells by competing for vital nutrients
(3) Never exhibit metastasis
(4) Regulated and controlled cell division

158. Find the correct statement

- (1) T-cells secrete antibodies and also help B-cells to produce them
(2) Innate immunity is pathogen specific and is based on memory
(3) Antibodies provide humoral immune response
(4) *Streptococcus pneumoniae* infects mainly kidney

159. Choose the statement not suitable for T-lymphocytes

- (1) Bone marrow and thymus both provide micro environments for the development and maturation of T-lymphocyte
(2) T-cells mediate cell mediated immunity
(3) T-lymphocytes secrete cytokines
(4) T-cells make clones consists of helper T-cells, killer-T cells and plasma cells

160. Which of the following is incorrect match regarding the drug and its source?

- | Drug | Source |
|--------------------------------------|---|
| (1) LSD (Lysergic acid diethylamide) | Fruiting bodies of the fungus <i>Claviceps purpurea</i> |
| (2) Opium | Latex of unripe capsule of <i>Papaver somniferum</i> |
| (3) Hashish | Dried leaves of <i>Erythroxylum coca</i> |
| (4) Ganja | Dried unfertilised female inflorescence of hemp plant |

161. Which of the following imaging technique uses strong magnetic fields to accurately detect pathological and physiological changes in the living tissue and detect cancer?

- (1) MRI
(2) Sonography
(4) CT scanning
(4) Radiography

- 162.** Increased lipid synthesis by liver in alcoholics gives rise to
- (1) Achalasia
 - (2) Peptic ulcer
 - (3) Osteoporosis
 - (4) Fatty liver syndrome
- 163.** Following are the steps in MOET programme for herd improvement in which a cow has been administered hormones with FSH activity. What is the **correct** sequence of steps from A to D?
- A. Instead of one egg which they normally produce per cycle, it produces 6–8 eggs.
 - B. Fertilised eggs at 8-32 cells stage are recovered non-surgically
 - C. It is either mated with elite bull or artificially inseminated
 - D. Embryoes are transferred to surrogate mothers
- (1) A → C → D → B
 - (2) C → A → D → B
 - (3) C → A → B → D
 - (4) A → C → B → D
- 164.** Miller made the first successful simulation experiment to assess the validity of the claim for origin of organic molecules in the earlier earth conditions. All the given statements are correct w.r.t Miller's experiment except
- (1) Mixture is heated to 800°C
 - (2) He found formation of amino acids
 - (3) He circulated the mixture under vacuum condition
 - (4) He used a mixture of H₂O, CH₄, NH₃, H₂ and CO₂ for his experiment
- 165.** Select the correct statement
- (1) Analogous organs show divergent evolution
 - (2) Homologous organs have different fundamental structure but common function
 - (3) Thorn of *Bougainvillea* and Tendril of *Cucurbita* are homologous structure
 - (4) Wings of butterfly and birds are homologous to each other
- 166.** Select the **correct** statement w.r.t. Lederberg's replica experiment
- (1) Mutations appears only after the exposure to the suitable environment
 - (2) The presence of penicillin causes change in DNA replication pattern that results in mutation
 - (3) Mutations occurred in the large numbers of bacteria due to penicillin
 - (4) The new environment only selects the preadaptive mutations that occurred earlier
- 167.** Hardy-Weinberg equilibrium law states that the relative frequency of alleles in the population of sexually reproducing organisms remains constant from generation to generation if
- a. The population is large enough
 - b. Random mating occurs
 - c. No mutation occurs
 - d. No migration occurs
- (1) a, b & c are correct
 - (2) a, b & d are correct
 - (3) a, b, c, d are correct
 - (4) b, c & d are correct
- 168.** Which of the following having highest cranial capacity?
- (1) *Homo habilis*
 - (2) *Homo erectus*
 - (3) *Homo sapiens neanderthalensis* (Neanderthal man)
 - (4) *Homo sapiens fossilis* (Cromagnon man)
- 169.** How many followings statements are correct ?
- A.** Hugo deVries brought forth the idea of mutations – large difference arising slowly and gradually in a population.
 - B.** Mutations are random and directionless while Darwinian variations are small and directional.
 - C.** Darwin believed mutation caused speciation and hence called it saltation (single step large mutation).
- (1) Three
 - (2) Two
 - (3) One
 - (4) None

170. Which of the followings statements are correct?

- A.** Disturbance in genetic equilibrium, or Hardy-Weinberg equilibrium, i.e., change of frequency of alleles in a population would be interpreted as resulting in evolution.
- B.** There would be a gene flow if gene migration, happens multiple times.
- C.** Natural selection is a process in which heritable variations enabling better survival are enabled to reproduce and leave greater number of progeny.

- (1) A and B
- (2) B and C
- (3) A and C
- (4) A, B and C

171. How many followings statements are correct ?

- A.** T-lymphocytes are part of acquired immunity.
- B.** Each antibody molecule has four pairs peptide chains, two pairs small called light chains and two pairs long called heavy chains.
- C.** Antibody-mediated immune response is responsible for graft rejection.

- (1) Three (2) Two
(3) Zero (4) One

172. Which measure would be particularly useful for prevention and control of alcohol and drug abuse among adolescents ?

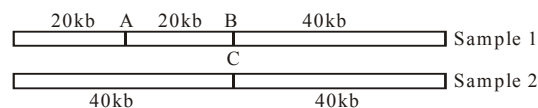
- Avoid undue peer pressure
- Seeking professional and medical help
- Looking for danger sign
- Education and counselling
- Seeking help from parents and peers

- (1) a, b, d
- (2) a, c, d, e
- (3) c, e
- (4) a, b, c, d, e

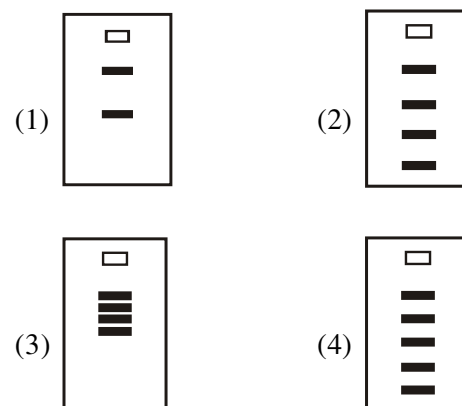
173. Which of the following is correctly matched :-

- (1) Gene gun method - suitable for plants
- (2) P.C.R. technique - cutting of D.N.A at specific location
- (3) Gel electrophoresis - Gene cloning
- (4) pBR322 - variety of virus

174. In the following diagram A,B and C are restriction sites and same DNA molecule is cut by restriction endonucleases



If both samples are loaded in a single well of gel after restriction digestion then find out correct band pattern in gel electrophoresis :-



175. A novel strategy was adopted to prevent nematode infestation in plant which was based on the process of RNA interference technique. Which of the following is not the part of this technique?

- (1) Involves silencing of specific mRNA due to complementary ds RNA molecule
- (2) Prevent the translation on the specific mRNA
- (3) Using *Salmonella typhimurium* nematode specific genes are introduced into the host plant.
- (4) Source of this complementary RNA could be from infection by viruses having RNA genomes or transposons

176. Which of the following statement is correct ?

- (1) *Bacillus thuringiensis* produces herbicidal protein
- (2) Bt toxin is encoded by a gene named *fry*.
- (3) Protein encoded by genes *cryI*Ac & *cry II*Ab control the round worms.
- (4) *Bt* toxins can specifically kill specific pest (insects).

177. 'Rosie' a transgenic cow is known to produce a type of milk, which has all the following characteristics except :-

- (1) Protein content of 2.4 g / litre
- (2) Has human α -lactalbumin
- (3) Can be used for treatment of cystic fibrosis
- (4) More balanced diet than normal cow milk for babies

178. Match the column-I with column-II :

Column-I	Column-II
(i) Golden rice	(A) Increased self life
(ii) Flavr savr tomato	(B) HGH
(iii) Mouse	(C) Vitamin A
(iv) Transgenic pig	(D) Organ transplantation
(1) (i) – C, (ii) – A, (iii) – B, (iv) – D	
(2) (i) – B, (ii) – A, (iii) – C, (iv) – D	
(3) (i) – B, (ii) – C, (iii) – A, (iv) – D	
(4) (i) – D, (ii) – A, (iii) – B, (iv) – C	

179. Alcoholic beverages having maximum & minimum alcoholic concentration respectively are :-

- (1) Wine & Beer
- (2) Beer & Brandy
- (3) Wine & Vodka
- (4) Brandy & Beer

180. Select the incorrect match:-

- (1) Large holes – Cottage cheese.
- (2) Streptokinase – Clot buster
- (3) Glomus – Mycorrhizal fungus
- (4) Methanogens – Biogas

Your moral duty is to prove that
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AIR-69  PRIYANSHU AGARWAL Classroom	AIR-70  ISHIKA KAMRA Classroom	AIR-71  STUTI KHANDWALA Classroom	AIR-72  NAVEEN SHARMA Distance	AIR-75  SARTHAK SHARMA Classroom	AIR-76  VANSHIKA ARORA Distance	AIR-79  SAIYAM V. BHATT Classroom	AIR-81  UDITH MOHAN Distance	AIR-84  PRAJNAPAN BASU Distance	AIR-85  HARSH RAMOLIYA Distance	AIR-86  UTKARSH SHARMA Classroom	AIR-87  NISHTHA Distance	AIR-89  ARYAN BANSAL Distance	AIR-91  SANGEETA AGRAWAL Classroom	AIR-92  YASHDEEP Classroom	AIR-93  ISHAN GUPTA Classroom	
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